Santu Yadav — Computer Engineer

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Profile

Recent Computer Engineering graduate from **Pokhara University** with strong focus on full-stack web development, AI/ML integration, and native Android application development using Kotlin.
Full-stack developer (MERN stack) and Android developer (Kotlin + XML) with hands-on experience in designing and implementing end-to-end academic projects, including the **Advanced Attendance System**, an AI/ML-enabled closed-system attendance platform developed as a final-year capstone project.
Experienced in building complete application stacks covering frontend, backend, authentication systems, and basic machine learning integration. Currently expanding skills in cloud deployment and scalable system architecture using AWS.

Technical Skills

Languages: JavaScript, Kotlin, C, C++, Node.js, Express.js

Web: HTML, CSS, Tailwind CSS, React.js, Vue.js

Databases: MongoDB (Mongoose ODM), PostgreSQL, SQL

Other: OOP Concepts, UX/UI Design, Git/GitHub, RESTful APIs, FastAPI, Integraion of AI/ML pretrained/custom trained models into WebApp.

Projects

1. Advanced Attendance System Using Facial Detection and Recognition

A 5-credit major capstone project completed as part of the **Bachelor of Computer Engineering** degree under **Pokhara University**, carrying one of the highest individual subject weightages within the 137-credit engineering curriculum. The project was successfully defended, and the team achieved the highest possible grade: **A Grade**.
Developed a secure, full-stack web application that automates attendance management using a custom-trained CNN-based facial detection and recognition model within a closed-system architecture. The platform restricts access exclusively to users with pre-generated institutional credentials, ensuring that unauthorized users cannot access the system or utilize its features.
Key Contributions:
- Designed and developed secure role-based dashboards for **Admin, Teacher, and Student** users using **React** and **Tailwind CSS**, ensuring a responsive and intuitive user experience.
- Implemented and integrated backend RESTful APIs for functionalities such as **Add Teacher**, **Add Classroom**, authentication, and attendance management using **Node.js** and **Express.js**.
- Architected the backend following the **MVC (Model–View–Controller)** design pattern to improve scalability, maintainability, and code organization.
- Developed and integrated a CNN-based facial detection and recognition pipeline for automated attendance verification and identity matching.
- Implemented secure authentication and controlled-access mechanisms to maintain a closed-system environment with institution-authorized access only.

2. Smart Invoice

A full-stack, web-based invoicing platform developed using the **MEVN stack**, designed to streamline invoice creation and management.
Key Contributions:
- Led front-end development, building responsive and user-friendly interfaces for invoice generation and management.
- Implemented core invoicing logic and automated **PDF generation** using libraries such as **jsPDF**.
- Contributed to database migration and cloud deployment by transitioning data storage to **MongoDB Atlas**.

3. Real-Time Accident Reporting System

A real-time, life-saving emergency reporting platform conceptualized to address the lack of immediate accident response systems in Nepal.

Key Contributions:

- Researched, designed, and developed the solution during a college hackathon to enable rapid accident reporting by bystanders.
- Integrated **Google Maps APIs** (Maps, Places, Distance Matrix) to identify and rank the nearest hospitals, ambulances, and police stations.
- Implemented **Twilio WhatsApp API** to send real-time alerts with precise location and incident details to emergency services.

4. Student Management System

An individual full-stack web application built using the **MEVN stack**, aimed at managing student records and academic workflows.

Status: Currently under development.

5. CEE Rank Predictor

An AI-powered, web-based rank prediction system for medical entrance aspirants, built using a custom-trained machine learning model.

Key Contributions:

- Trained a custom AI/ML model using real historical datasets from previous years' **Common Entrance Examinations (CEE)**.
- Developed a web interface to provide rank predictions and insights based on candidate performance.

6. Inventory Management System

A Windows-based desktop application developed in **C#** for managing inventory records and operations.

Key Contributions:

- Designed and implemented intuitive UI components to improve usability and data visualization.

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Achievements

ISTN Hackathon 2025 Winner: Led my team to first place at the 24-hour ISTN Hackathon 2025 (organized by IT Skills Training Nepal) with an idea I'd researched and developed over 2-3 months beforehand. Instead of scrambling during the event, I came prepared with a solid concept and technical direction, which allowed us to execute efficiently and win. The preparation really made the difference in how smoothly the hackathon went.

TechSprint Hackathon 2026 (Team Lead): Led the same team at the 48-hour TechSprint Hackathon 2026 (January 17-19, 2026) at Nepal Engineering College, organized by CoESS. We built on the momentum and collaboration from ISTN, bringing the same strong teamwork to a highly competitive national event. It was great seeing the team grow and handle bigger challenges together at bigger stage.

Participated in The Cosmos Hackathon as Mentor: Organized by ICT CLUB CCMT, PU.

Participated in Ambition Hack Fest 2025: A 48 hours national level hackathon organized by Ambition College, Mid-Baneshwor

GPA:: 3.83 in 1st semester and 3.64 in 8th semester (Pokhara University)

University Scholarship Holder: Got fully-funded scholarship from Pokhara University (2078 Intake of PU).

Pokhara University Master's Scholarship Entrance Examination *March 2083 B.S. (2026 A.D.)*
ME Computer / MSc Computer Science, March 2026 Intake — Faculty of Science & Technology. Qualified among **63 successful candidates** out of hundreds of applicants, scoring **80/150**, ranked **34$^{\text{th}}$ overall** and **28$^{\text{th}}$ among public-sector candidates**.

References

Dr. Ram Govind Aryal
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